

Stage a Table Centerpiece SOP (8x Pieces)

One episode is one full centerpiece staging. It runs from a crooked runner and seven loose pieces sitting on the staging strip, to a squared runner down the middle of the table with the vase at its middle, one candle out each side of the vase, and four coasters lying flat, two a side, outside the candles.

The two grippers have fixed sides and never swap. The **right gripper is the primary gripper**: it works the whole right half of the runner line, the right end of the runner, the vase, the right candle, and the two right coasters. The **left gripper is the support gripper**: it works the whole left half of the runner line, the left end of the runner, the left candle, and the two left coasters. The vase sits at the middle and is always placed by the right gripper. Both arms work together only where this SOP says so: the corner pinch that drags the runner back in Step 1, and the anchor that holds the coaster stack in Step 4. Everywhere else one arm works while the other waits, and nothing is ever passed from one gripper to the other.

The order never changes: **lay and square the runner, place the vase, place the two candles, place the four coasters, adjust the spacing, then square up and finish**. Nothing goes on the runner until the runner is square.

The runner is pulled along the table and never carried through the air. Never lay a piece on the bare table once the runner is down. Never place a piece before the runner is square. Never move more than one piece per trip. Never grip the vase by its rim. Never turn a piece in the air. Never let a piece hang over the runner edge or the table edge. Never let two pieces touch. Never work the other gripper's side.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera**: 900 mm.
- **cell_type**: bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the table and its frame includes the full runner line from end to end, both halves of it, the whole staging strip, and the front table edge.
3. Both arms are at home with grippers open.
4. The table holds nothing but the runner, the vase, the two candles, and the coaster stack.

5. The right arm reaches the right end of the runner, the whole right half of the runner line, the vase, the candle staging spot, and the coaster stack without extending to a joint limit. The left arm reaches the left end of the runner, the whole left half, and the candle staging spot the same way. Shorten the runner line until all of that is true.
6. Each arm reaches its own half from above and from the front, and neither arm has to reach across into the other half.
7. Arm paths between the staging strip and the runner line are clear.

Materials checklist

1. The runner lies open on the staging strip, in front of the runner line. It is not folded and not rolled.
2. The runner lies clearly crooked and off center. The mess is obvious at a glance.
3. The runner line is clear, and the runner fits on it with nothing hanging over a table edge.
4. The vase stands upright on its own on the staging strip. It is empty and dry, has a round body, and can be lifted by one gripper.
5. Two identical thick pillar candles stand upright on their own at the candle staging spot, not touching each other or any other piece. Both candles start at this one spot every episode, whichever half they end up on.
6. Four coasters sit on the staging strip in one stack of four, square and tidy, so the top coaster can be gripped by its edge.
7. Nothing at all stands on the runner line.

Workspace layout

Nothing on the table is marked or taped out. Every area below is a piece of the bare tabletop, judged by eye against the table edges and the pieces themselves.

- **Front table edge:** the table edge nearest the station. Every straight line in this task is judged against it. Nothing may hang over it.
- **Runner line:** the strip of table down the middle, running left to right, where the runner ends up. It is empty at episode start and holds the finished centerpiece at episode end.
- **Right half:** the runner line from its middle to its right end. Worked by the right gripper only.
- **Left half:** the runner line from its middle to its left end. Worked by the left gripper only.
- **Staging strip:** the clear strip of table between the runner line and the front table edge. It holds the runner, the vase, the two candles, and the coaster stack at episode start, and it is empty at episode end.
- **Candle staging spot:** the one place both candles start, at the front of the staging strip, just left of the middle of the table. The two candles stand together there, clear of each other. Both grippers pick their candle from this same spot, and it is empty at episode end.
- **Arm home:** the resting pose for each arm, gripper open and clear of the table.

Vocabulary

- **Arm assignment:** fixed by side, not by what is easiest to reach. The **right gripper** works the right half, the right end of the runner, the vase, the right candle, and the two right coasters. The **left gripper** works the left half, the left end of the runner, the left candle, and the two left coasters. No piece is ever passed from one gripper to the other.
- **Idle arm:** the arm not named in a step and not anchoring. An empty idle arm waits at home, clear of the table. Outside the corner pinch in Step 1 and the anchor in Step 4, it never steadies a piece, holds the runner down, or helps the working arm.
- **Piece:** any one of the eight things staged in this task: the runner, the vase, either candle, or any coaster.
- **Right candle and left candle:** the two candles are identical and both start at the candle staging spot. The candle the right gripper takes is the right candle, and the one the left gripper takes is the left candle. The names say which half a candle ends on, not where it starts.
- **Runner:** the long open cloth that goes down the middle of the table. It is pulled along the table, not carried through the air.
- **Drag:** a gripper closes lightly on the cloth and pulls it a short way along the table, then opens. The cloth stays low and against the table. A pinched corner coming up a little is expected and is not a problem.
- **Corner pinch:** both grippers close lightly on the two near corners of the runner at the same time and pull it back together. This is the two arm move that lays the runner in Step 1.
- **Anchor:** a gripper holds the coaster stack down on the staging strip, without lifting it, while the other gripper takes the top coaster off it. An anchoring gripper never takes a coaster itself.
- **Nudge:** a gripper closes lightly on a piece and moves it a short way, then opens. The piece may come off the table as the gripper pinches it, and lifting it clear and setting it back down is fine where that places it better. A nudge never carries a piece onto the other half.
- **Runner flat:** the whole runner lies in one layer on the table. No part of it is bunched, folded over, or doubled back on itself.
- **Runner square:** the long edge of the runner runs parallel to the front table edge along its full length, and the runner is flat.
- **Middle of the runner:** the point that looks the same distance from each end of the runner, judged by eye against the two ends.
- **Coaster stack:** the four coasters standing in one pile on the staging strip. Coasters are always taken one at a time off the top, gripped by the edge.
- **Coaster flat:** the whole face of the coaster lies on the runner. It does not stand on edge, lean, ride on another coaster, or sit on a part of the runner that hangs off the runner line.

- **Narrow gap:** a clear space between two pieces, small enough to read as a pair but wide enough that the two pieces plainly do not touch.
- **Halves read the same:** the right half and the left half look like copies of each other. The candle looks the same distance out from the vase on each side, and each pair of coasters looks the same distance out from its candle.
- **Stable placement:** the piece stays still after the gripper opens, sits fully on the runner, and does not rock, lean on a neighbour, or slide.
- **Release point:** a gripper opens only over the staging strip or over the runner. Never over another piece and never over bare table once the runner is down.

Steps

Step 1: Lay and square the runner

Goal: the runner lies open and square along the runner line, ready for pieces to go on it.

Both arms work together in this step. They pinch the two near corners of the runner and drag it back onto the runner line as one move, and the squaring happens in that same drag.

- Each gripper comes down on the near corner of the runner on its own side, the **right gripper** on the right corner and the **left gripper** on the left corner, and closes lightly on the cloth.
- Both arms pull back together, evenly and at the same speed, dragging the runner onto the runner line. An even pull brings the runner back straight instead of turning it.
- Keep pulling the corners apart just enough to take the slack out, so the long edge comes back lining up with the front table edge and any corner tucked under the runner is pulled out flat as the runner moves.
- Open both grippers and lift clear once the runner lies on the runner line in one layer.
- The runner stays low and against the table the whole way. It is never carried through the air.

Squaring the runner. The corner pinch usually leaves the runner square. Whatever is left is corrected one half at a time, the right half first, then the left half, each gripper working only its own half.

- If a half is short of its end of the runner line, still bunched, or still tucked under, the gripper on that half closes on the runner and drags it, then opens and lifts clear.
- If the long edge on a half sits at an angle to the front table edge, the gripper on that half presses lightly on that edge and drags it a short way. Stop when that edge runs parallel to the front table edge and the cloth on that half is flat.
- Use two small drags at most on a half. The other arm is the idle arm while one half is corrected.

Check: the runner lies open along the runner line in one layer, its long edge runs parallel to the front table edge along its full length, no part of it is bunched, folded over, doubled back, or tucked under itself, and no part of it is left on the staging strip. If one half is off, give it one more small drag with the gripper on that half. If the runner still will not lie open and square, end the episode and fix the runner in the reset. Nothing goes on the runner until this check passes.

Step 2: Place the vase

Goal: the vase stands upright at the middle of the runner.

- The **right gripper** comes down on the vase from above and closes on its body, below the rim, so the vase hangs upright and not tilted.
- If the grip is not solid before the lift, open and try again. Two tries at most. If the vase still will not hold, end the episode and check the vase in the reset.
- Lift the vase straight up clear of the staging strip and carry it level to the runner.
- Stand it at the middle of the runner, judged by eye against the two ends.
- Open the gripper once the vase stands on its own, fully on the runner, and touches nothing.

The left arm is the idle arm for the whole of Step 2.

Check: the vase stands upright at the middle of the runner and looks the same distance from each end. It sits fully on the runner and hangs over nothing. If it is off the middle, tilted, or part off the runner, lift it clear with the right gripper and set it down again once.

Step 3: Place the two candles

Goal: one candle stands on each half of the runner, out from the vase.

Both candles start at the candle staging spot. One candle per trip. Finish the right candle before starting the left one.

- The **right gripper** closes on the body of either candle at the candle staging spot, lifts it straight up off the staging strip, and carries it level to the right half.
- Stand it on the runner between the vase and the right end of the runner.
- Open the gripper once the candle stands on its own, fully on the runner, and touches nothing.
- The **left gripper** does the same with the candle still standing at the candle staging spot, on the left half, between the vase and the left end of the runner.
- Never take both candles on one trip.
- Never turn a candle in the air, and never set a candle on the bare table.

Each arm is the idle arm while the other one places its candle.

Check: both candles stand upright on the runner, one per half, and each candle looks the same distance out from the vase. Nothing touches. If a candle leans, sits part off the runner, or is

plainly further out than the other one, lift it clear with the gripper on that half and set it down again once.

Step 4: Place the four coasters

Goal: four coasters lie flat on the runner, two per half, outside the candles.

One coaster per trip. Do the two right coasters first, then the two left coasters. Both arms work in this step: the arm that is not carrying a coaster anchors the stack, so the top coaster can be picked off cleanly instead of dragging the stack with it.

- The **left gripper** anchors the coaster stack, holding it down on the staging strip without lifting it, and holds that anchor for both right coasters.
- With the stack anchored, the **right gripper** closes on the edge of the top coaster and lifts it straight up clear of the stack.
- Lay it flat on the right half, outside the right candle, and open the gripper.
- Take the next top coaster the same way and lay it flat outside the first one, with a narrow gap between them, then the left gripper releases the anchor.
- The arms now swap roles. The **right gripper** anchors the stack the same way, and the **left gripper** takes the top coaster off it, lays it flat on the left half outside the left candle, and repeats for the fourth coaster.
- An anchoring gripper only holds the stack still. It never lifts the stack, never takes a coaster itself, and never touches a coaster already on the runner.
- Every coaster lies flat. Never lean one on edge, never let two overlap, and never lay one on a part of the runner that hangs off the runner line.

Neither arm is idle in this step. One picks and lays, the other anchors.

Check: four coasters lie flat on the runner, two per half, all of them outside the candles, none overlapping and none touching. The two on the right look like the two on the left. If a coaster is on edge, overlapping, or off the runner, lift it clear with the gripper on that half and lay it down again once.

Step 5: Adjust the spacing

Goal: the right half and the left half read the same.

Work the right half first, then the left half, and each gripper works only its own half.

- The vase stays where it is. It marks the middle, so every gap is set by moving the candle or a coaster.
- With the **right gripper**, close lightly on the right candle and move it along the runner until the gap between the vase and the candle looks right, then open. The piece coming off the table as the gripper pinches it is fine, and lifting it clear and setting it straight back down is fine where that places it better.

- Move each right coaster the same way, so the gap between the candle and the first coaster and the gap between the two coasters both look even.
- With the **left gripper**, set the left candle and the two left coasters the same way, matching what the right half now looks like.
- Use two small moves per piece at most. A piece is never carried onto the other half and never turned in the air.

Each arm is the idle arm while the other one adjusts its half.

Check: the two candles look the same distance out from the vase, each pair of coasters looks the same distance out from its candle, no two pieces touch, and nothing hangs over the runner edge or the table edge. If one half is out, give the piece one more small nudge with the gripper on that half.

Step 6: Square up and end the episode

- Confirm the runner: its long edge runs parallel to the front table edge, and no part of it is bunched or folded over.
- Confirm the vase: upright at the middle of the runner.
- Confirm the candles: one per half, each looking the same distance out from the vase.
- Confirm the coasters: four lying flat, two per half, outside the candles, and evenly out.
- Confirm the gaps: no two pieces touch, nothing hangs over the runner edge, nothing hangs over a table edge, and the staging strip is empty.
- Fix a failed check only with the fix written in its step. If no fix is written, or the fix would mean going back through the order, end the episode and reset instead.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the four coasters off the runner and rebuild them as one square stack of four on the staging strip, with the top coaster free to be gripped by its edge.
2. Stand the two candles at the candle staging spot, upright and clear of each other and of every other piece.
3. Stand the vase on the staging strip, upright, empty and dry.
4. Drag the runner off the runner line onto the staging strip and leave it open, crooked, and off center, with no part of it on the runner line.
5. Replace any piece that will not stand upright on its own, and replace a runner that will not lie open in one layer.
6. Wipe the runner line and the staging strip and clear any bits off the table.
7. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Runner left bunched or folded over

- **Visible cue:** the finished runner shows a fold, a ridge, a doubled back end, or a bunched patch, or a part of it is still on the staging strip after the left and right grippers finish.
- **SOP rule broken:** Step 1, the runner ends open in one layer with nothing bunched, folded over, or doubled back.
- **Coaching note:** read the whole runner end to end before you go on. Drag the bad part open with the gripper on that half.

Violation: Runner not squared before pieces go on

- **Visible cue:** the vase, a candle, or a coaster goes down while the runner still sits at an angle to the front table edge, or a gripper squares the runner after pieces are already on it.
- **SOP rule broken:** Steps 1 and 2, the runner is square and flat before the right gripper places the vase.
- **Coaching note:** square first, place second. Squaring a loaded runner drags everything with it.

Violation: Vase gripped by its rim

- **Visible cue:** the right gripper closes on the rim or the top opening of the vase instead of on its body.
- **SOP rule broken:** Step 2, the right gripper closes on the vase body, below the rim.
- **Coaching note:** grip the body. A vase held by the rim tips as it lifts.

Violation: Vase not at the middle of the runner

- **Visible cue:** the vase comes to rest plainly nearer one end of the runner than the other, off the runner, or leaning.

- **SOP rule broken:** Step 2, the right gripper stands the vase at the middle of the runner, judged by eye against the two ends.
- **Coaching note:** look at both ends of the runner before you open the gripper.

Violation: Candle taken from the wrong place

- **Visible cue:** a candle starts the episode somewhere other than the candle staging spot, or a gripper picks a candle up from anywhere else on the staging strip.
- **SOP rule broken:** Step 3 and the materials checklist, both candles start at the candle staging spot and both grippers pick from that spot.
- **Coaching note:** both candles start in the same place every episode, and that is the only place a candle is picked up from.

Violation: Coaster not laid flat

- **Visible cue:** a coaster stands on edge, leans, rides on another coaster, or sits on a part of the runner that hangs off the runner line, after the left or right gripper releases it.
- **SOP rule broken:** Step 4, every coaster lies flat on the runner with none overlapping.
- **Coaching note:** take the coaster by its edge and lay the whole face down before you open.

Violation: Coasters in the wrong place or wrong number

- **Visible cue:** a half ends with more or fewer than two coasters, a coaster sits between the vase and a candle instead of outside the candle, or a gripper takes a coaster from anywhere but the top of the stack.
- **SOP rule broken:** Step 4, two coasters per half, taken one at a time off the top of the stack and laid outside the candle on that half.
- **Coaching note:** two a side, always outside the candle, always off the top of the stack.

Violation: Coaster stack not anchored

- **Visible cue:** a gripper takes a coaster off the stack while the other arm is at home or somewhere else, the anchoring gripper lifts the stack off the staging strip, the anchoring gripper takes a coaster itself, or the stack slides or topples during a pick.
- **SOP rule broken:** Step 4, one gripper holds the stack down while the other takes the top coaster off it, and the arms swap roles for the left half.
- **Coaching note:** anchor first, then pick. Hold the stack still and take one coaster off the top.

Violation: Vase moved during the spacing adjust

- **Visible cue:** a gripper moves the vase in Step 5 instead of setting the gaps with the candles and coasters.
- **SOP rule broken:** Step 5, the vase marks the middle and stays where it was placed.

- **Coaching note:** the vase is the reference. Set every gap by moving the candle or the coaster.

Violation: Halves do not read the same

- **Visible cue:** at the end, one candle sits plainly further out from the vase than the other, or one pair of coasters sits plainly further out or wider apart than the other pair.
- **SOP rule broken:** Steps 3, 4, and 5, each candle sits the same distance out from the vase and each pair of coasters sits the same distance out from its candle.
- **Coaching note:** set the right half, then copy it on the left. Read both halves together before you finish.

Violation: Piece off the runner, touching, or hanging over

- **Visible cue:** a piece comes to rest on the bare table instead of the runner, two pieces stand against each other, a piece sits part on and part off the runner, or a piece hangs over the runner edge or a table edge.
- **SOP rule broken:** Steps 2 to 5, every piece ends fully on the runner, touching nothing, and hanging over no edge.
- **Coaching note:** check the landing spot before you open the gripper. Bare table is never a place for a piece.

Violation: More than one piece moved in one trip

- **Visible cue:** a gripper lifts, carries, or moves two pieces together, or lifts two coasters off the stack at once. Both grippers on the one runner in Step 1 is one piece, not two.
- **SOP rule broken:** Steps 2 to 5, one piece per trip.
- **Coaching note:** exactly one piece per trip, every time.

Violation: Wrong gripper or sides swapped

- **Visible cue:** the left gripper touches the vase, the right end of the runner, the right candle, or a right coaster; or the right gripper works the left half, the left end of the runner, or the left candle. Either gripper on its own corner in Step 1, and either gripper anchoring the coaster stack in Step 4, is correct and is not this violation.
- **SOP rule broken:** Steps 1 to 5, the right gripper works the right half and the vase, and the left gripper works the left half.
- **Coaching note:** each gripper stays on its own side. The vase always belongs to the right gripper.

Violation: Idle arm gets in the way

- **Visible cue:** outside the corner pinch in Step 1 and the anchor in Step 4, the arm not named in a step comes in, holds the runner down, steadies a piece, or helps the working arm. Both arms working together in Step 1 and Step 4 is correct and is not this violation.

- **SOP rule broken:** Steps 2, 3, 5 and 6, one arm works while the other waits at home.
- **Coaching note:** the two arms share the runner drag and the coaster anchor. Everywhere else, one works and the other waits at home.

Violation: Piece turned in the air or passed between grippers

- **Visible cue:** a gripper rotates the vase, a candle, or a coaster while it is off the table, or a piece goes from one gripper to the other.
- **SOP rule broken:** Steps 2 to 4, each piece is lifted, carried level, and set down by one gripper without being turned in the air.
- **Coaching note:** carry it as you picked it up. Nothing is handed over and nothing is spun mid carry.

Violation: Piece dropped, knocked over, or hit

- **Visible cue:** the vase, a candle, or a coaster falls or topples, an arm knocks a placed piece on the way in or out, or a gripper drags the runner out from under pieces already on it.
- **SOP rule broken:** Steps 1 to 5, move in and out clear and open the gripper only on a stable placement.
- **Coaching note:** check the path and the landing spot before you move, and come in from above and from the front.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a failed check not fixed, a crooked or bunched runner, a piece missing or off the runner, two pieces touching, something left on the staging strip, an arm near home but not at it, or a gripper not fully open.
- **SOP rule broken:** Step 6, confirm the runner, the vase, the candles, the coasters, and the gaps, then return both arms home with grippers open, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Faulty runner:** a runner that is torn, stiff, or so creased that a correct corner pinch and drag leaves a fold in it. Replace it before the next episode.
- **Faulty piece:** a vase, candle, or coaster that will not stand or lie the way this task needs, or that slips out of a correct grip. Replace it before the next episode.

- **An area shifts during the episode**, or the runner line turns out to sit outside an arm's comfortable reach, so an end of the runner or a piece cannot be reached without stretching or folding the arm.

Annotation subtasks (from SOP)

1. Pinch both corners and drag the runner back onto the runner line
2. Correct one half of the runner with a drag
3. Place the vase at the middle of the runner
4. Place one candle out from the vase
5. Anchor the stack and lay one coaster flat outside a candle
6. Move one piece to set its spacing
7. Return both arms home and end the episode